PHYSICS

21-05-2023

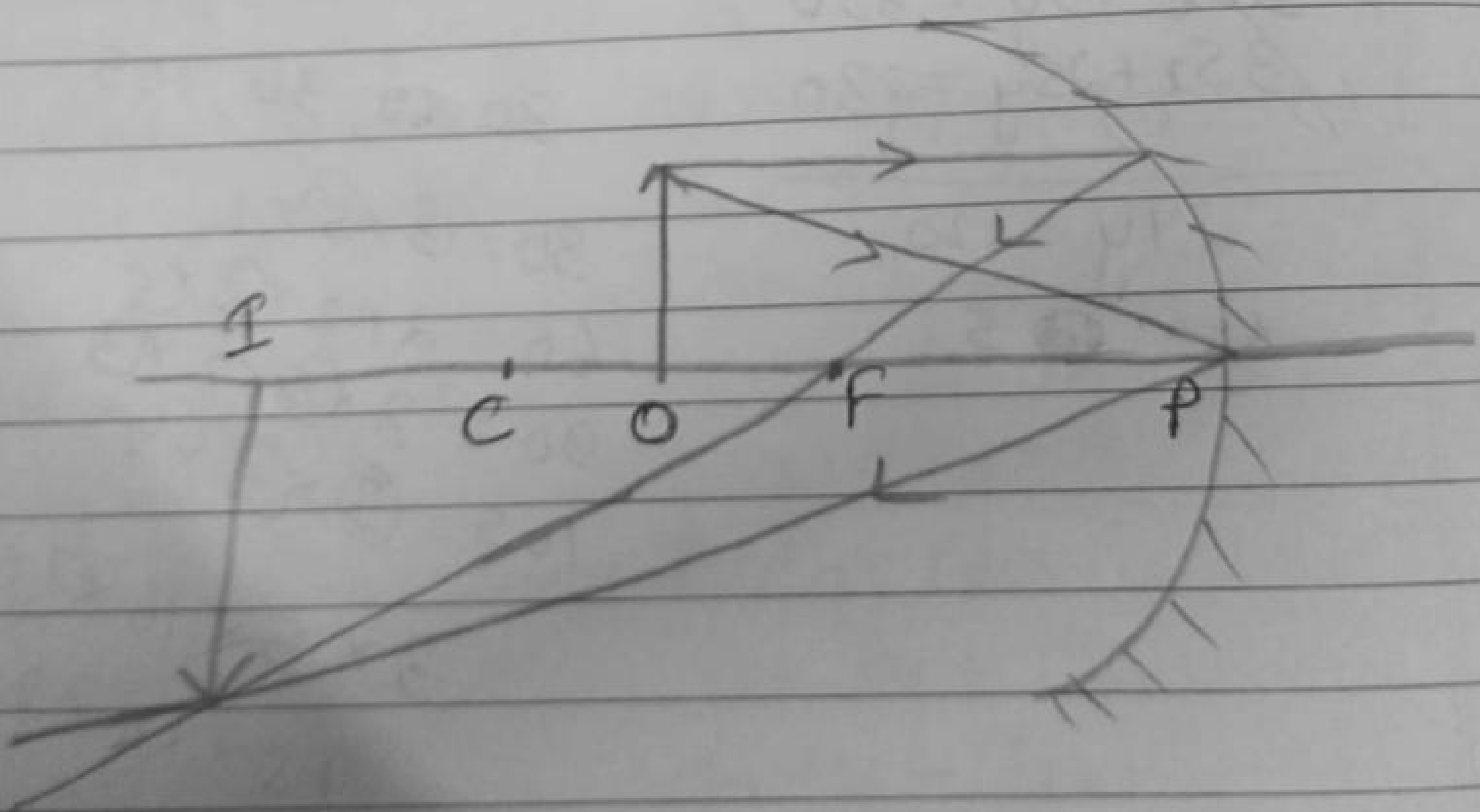
1. Parallel beam of light

2. $f = \frac{R}{2}$

$\Rightarrow f = 25\ cm$

It will provide convex mirror as ~~its~~ outward bulging surface of the sphere is to be used as mirror.

3.



When the object is placed between C and F the image will be formed beyond C and would real, enlarged and inverted.

4. height of object = 2cm
$f = 10cm$
$u = 15cm$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$= \frac{1}{10} = \frac{1}{15} + \frac{1}{v}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{10} - \frac{1}{15}$$

$$\Rightarrow \frac{1}{v} = \frac{3-2}{30} = \frac{1}{30}$$

$\therefore v = 30 cm$

Image is formed at a distance of 30cm from the mirror.

$$m = -\frac{v}{u} = \frac{-30}{15} = -2$$

The image of the is diminished and it real and inverted.